INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, ALLAHABAD

DATA COMMUNICATION AND NETWORKING LAB RECORD

EXPERIMENT - 4

GROUP 14

-IEC2023071 Aditiya Krishna Chauhan

Static IP Routing in Cisco Packet Tracer

Aim

To design and configure a multi-router network in Cisco Packet Tracer using **static routing** and verify communication between hosts in different IP networks.

Software Used

- Cisco Packet Tracer

Theory

Static routing is a fundamental routing technique in computer networks where routes are **manually configured** by a network administrator. Unlike dynamic routing protocols such as RIP (Routing Information Protocol) or OSPF (Open Shortest Path First), static routing does not involve any automatic exchange of routing information between routers. This makes it simple, predictable, and resource-efficient, but less flexible for large or frequently changing networks.

Every router maintains a **routing table**, which contains information about destination networks and the corresponding paths (next hops) to reach them. In static routing, entries are explicitly added to this table using commands like:

```
ip route <destination_network> <subnet_mask>
```

<next_hop_ip> Here:

- *Destination network* is the target network we want to reach.
- *Subnet mask* defines the size of that network.
- *Next-hop IP* is the IP address of the neighboring router that will forward the packet closer to the destination.

When a router receives a packet, it performs the following steps:

1. It examines the destination IP address of the packet.
2. It searches its routing table for a matching network.

3. If a match is found, it forwards the packet to the specified next-hop router.
4. If no match exists, the packet is dropped (unless a default route is configured).

In a static routing environment, **each router only knows about directly connected networks by default**. Therefore, routes to all other (remote) networks must be manually configured. This requires careful planning to ensure that all routers have complete and consistent routing information.

In this experiment, a **multi-router topology** was implemented using three routers connected in a linear (chain) fashion. Each router was connected to a local LAN through a switch, forming three separate subnetworks. The routers were interconnected using serial links, creating additional networks between them.

Static routes were configured on each router so that:

- Router0 could reach networks beyond Router1.

- Router1 could act as an intermediate router, forwarding traffic between the left and right networks.
- Router3 could reach networks on the left side via Router1.

This setup demonstrates how static routing enables **end-to-end communication across multiple networks** by explicitly defining the path that packets must follow. It also highlights key characteristics of static routing:

- **Deterministic behavior**: Routes do not change unless manually modified.
- **Low overhead**: No bandwidth or CPU is used for routing updates.
- **Limited scalability**: Configuration becomes complex as the number of networks increases.

Thus, static routing is best suited for small, stable networks or scenarios where precise control over routing paths is required.

Network Topology and Addressing

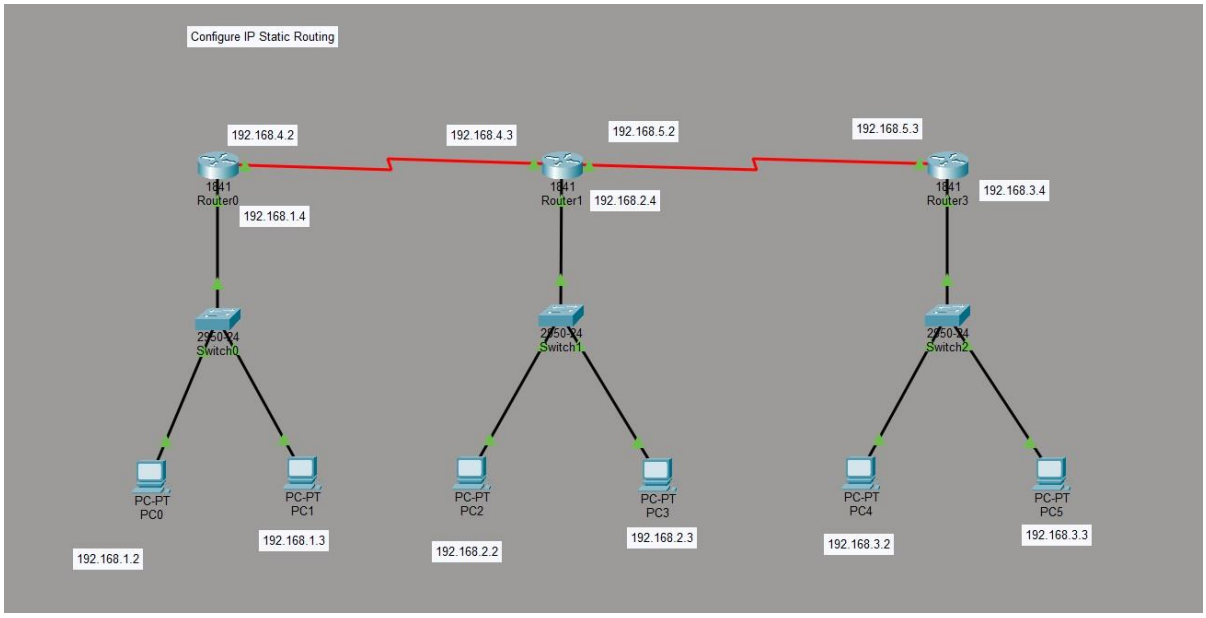

Figure 1. Network Topology

The following networks were used in the implementation:

- Left LAN: 192.168.1.0/24
- Middle LAN: 192.168.2.0/24

- Right LAN: 192.168.3.0/24
- Router0–Router1 link: 192.168.4.0/24
- Router1–Router3 link: 192.168.5.0/24

Example interface assignments observed in the topology:

- Router0 LAN interface: 192.168.1.4
- Router0 to Router1: 192.168.4.2

- Router1 to Router0: 192.168.4.3
- Router1 to Router3: 192.168.5.2
- Router3 to Router1: 192.168.5.3
- Router1 LAN interface: 192.168.2.4
- Router3 LAN interface: 192.168.3.4

PCs were assigned addresses in their respective LANs, such as 192.168.1.2, 192.168.1.3, 192.168.2.2, 192.168.2.3, 192.168.3.2, and 192.168.3.3.

Procedure

1. Open Cisco Packet Tracer and place three routers, three switches, and six PCs.
2. Connect each router to a switch using Ethernet links.
3. Connect the routers using serial links to form a chain topology.
4. Assign IP addresses to all router interfaces according to the selected addressing scheme.
5. Configure PCs with IP addresses, subnet masks, and default gateways matching their local router interfaces.
6. Configure static routes on each router so that packets can reach remote networks.
7. Verify the routing table on each router.
8. Test connectivity using ping and simulations between PCs in different LANs.

Static Routes Configured

Rout
er0

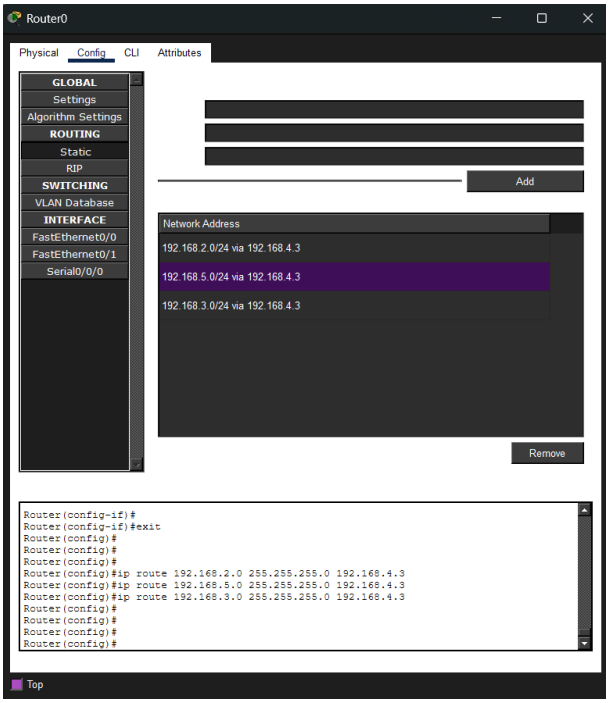

Figure 2. Router0 Routing Table

Routes were added toward the middle and right-side networks through Router1:

ip route 192.168.2.0 255.255.255.0 192.168.4.3

ip route 192.168.3.0 255.255.255.0 192.168.4.3

ip route 192.168.5.0 255.255.255.0 192.168.4.3

Router1

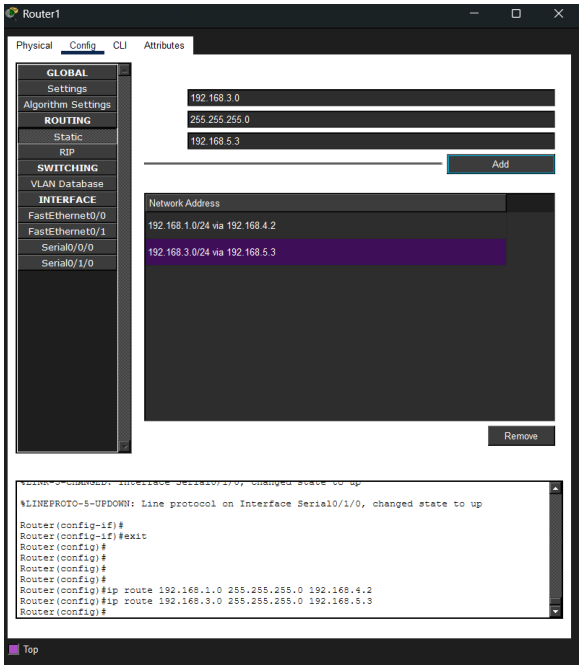

Figure 3. Router1 Routing Table

Routes were added toward the left LAN and right LAN:

ip route 192.168.1.0 255.255.255.0 192.168.4.2

ip route 192.168.3.0 255.255.255.0 192.168.5.3

**Rout
er3**

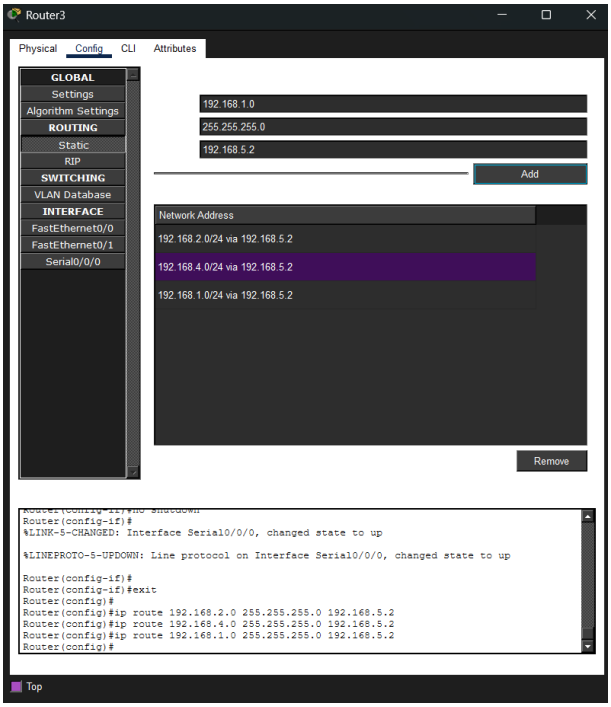

Figure 4. Router3 Routing Table

Routes were added toward the left-side networks through Router1:

ip route 192.168.1.0 255.255.255.0 192.168.5.2

ip route 192.168.2.0 255.255.255.0 192.168.5.2

ip route 192.168.4.0 255.255.255.0 192.168.5.2

Observations and Discussions

1. Behavior Before Configuring Static Routing

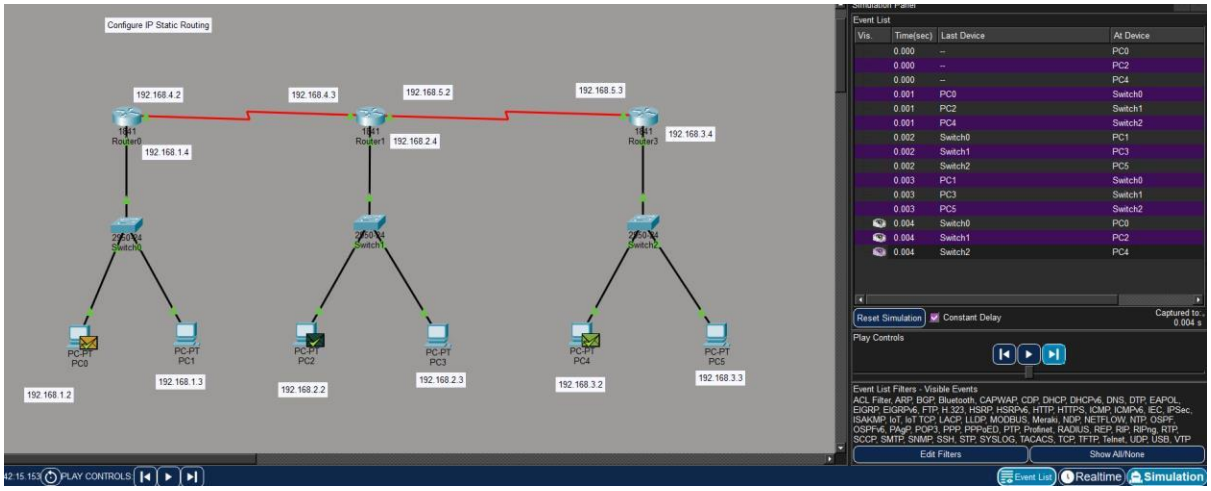

Figure 5. Packet transfer successful within same LAN before static routing

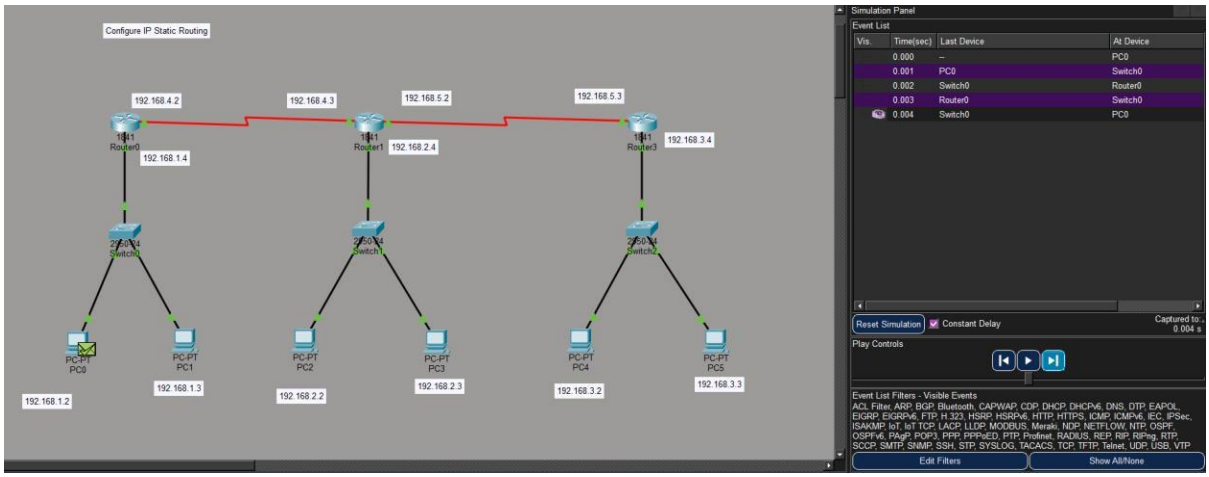

Figure 6. Packet transfer successful between PC0 and Router0 (same LAN) before static routing

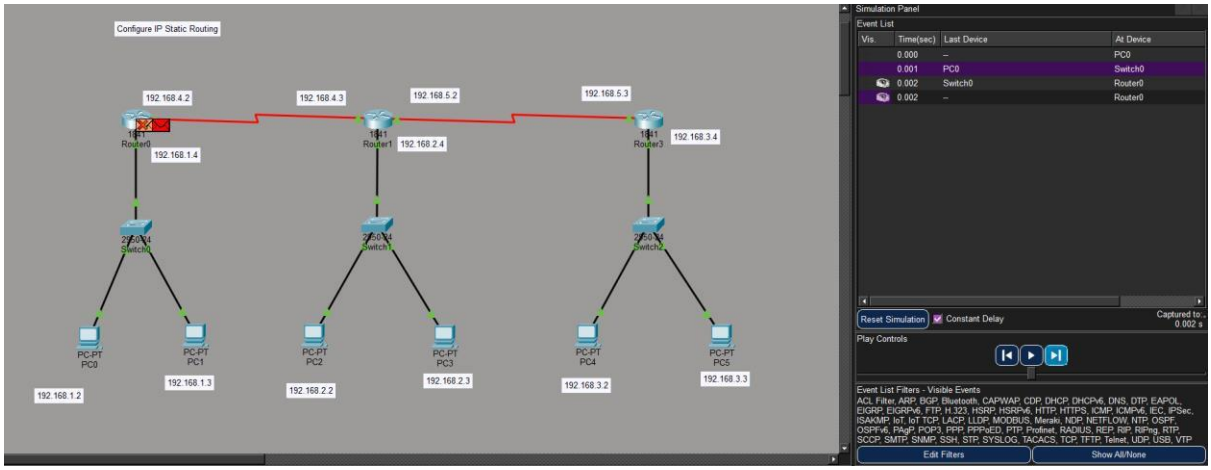

Figure 7. Packet transfer outside LAN failed at Router without static routing

- In simulation mode, packets generated from PCs in one LAN (e.g., 192.168.1.0/24) were able to reach only their **local switch and router**, but failed to proceed further toward remote networks.
- The event list showed packets moving from **PC → Switch → Router**, after which they were **dropped at the router**.
- This occurred because routers (Router0, Router1, Router3) only had knowledge of their **directly connected networks** and no information about remote networks.

- As a result, when a router received a packet destined for an unknown network, it had **no matching route in its routing table**, leading to packet drop.
- No successful end-to-end communication was observed between PCs belonging to different subnets.

2. Behavior After Configuring Static Routing

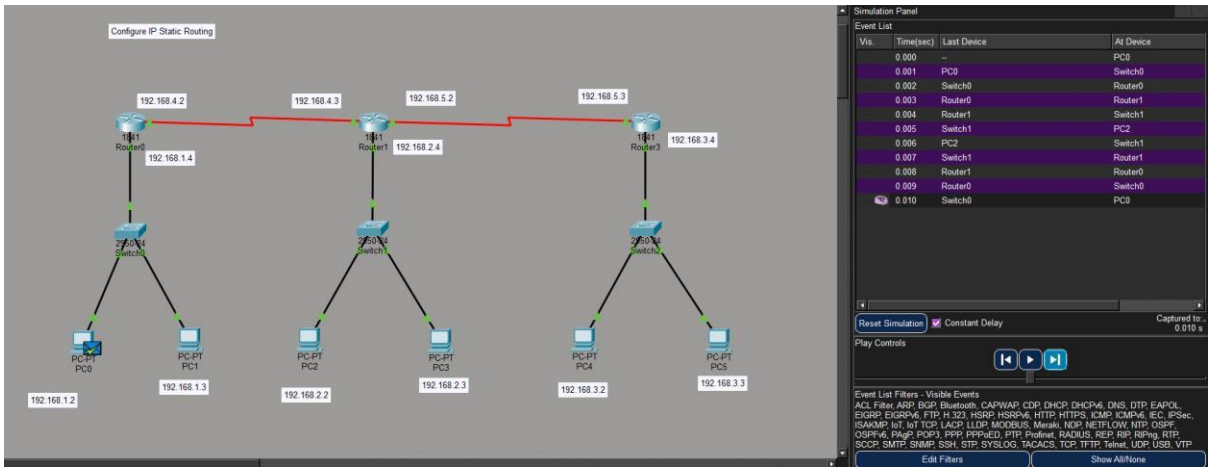

Figure 8. Packet transfer between 2 different LANs (1 and 2) successful after static routing

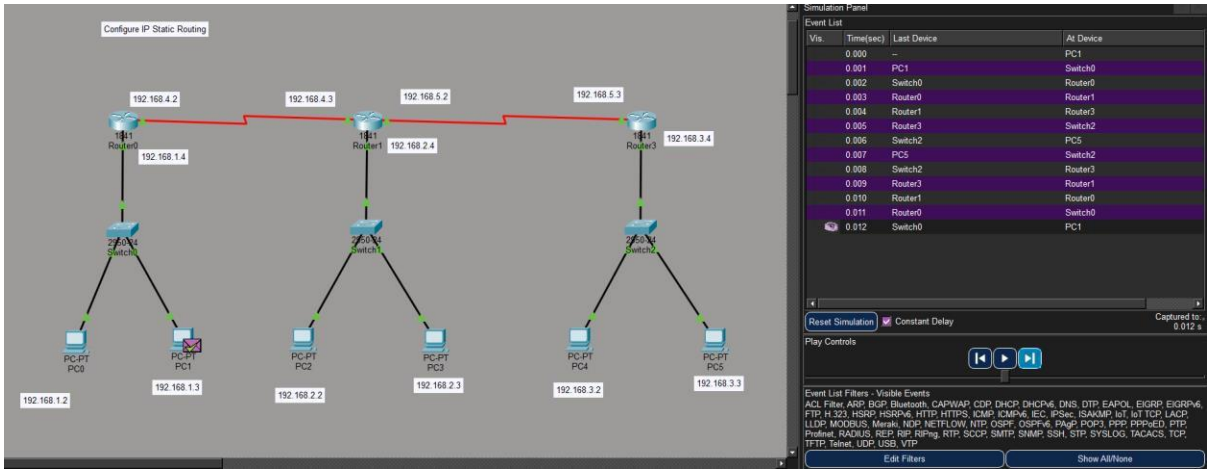

Figure 9. Packet transfer between 2 different LANs (1 and 3) successful after static routing

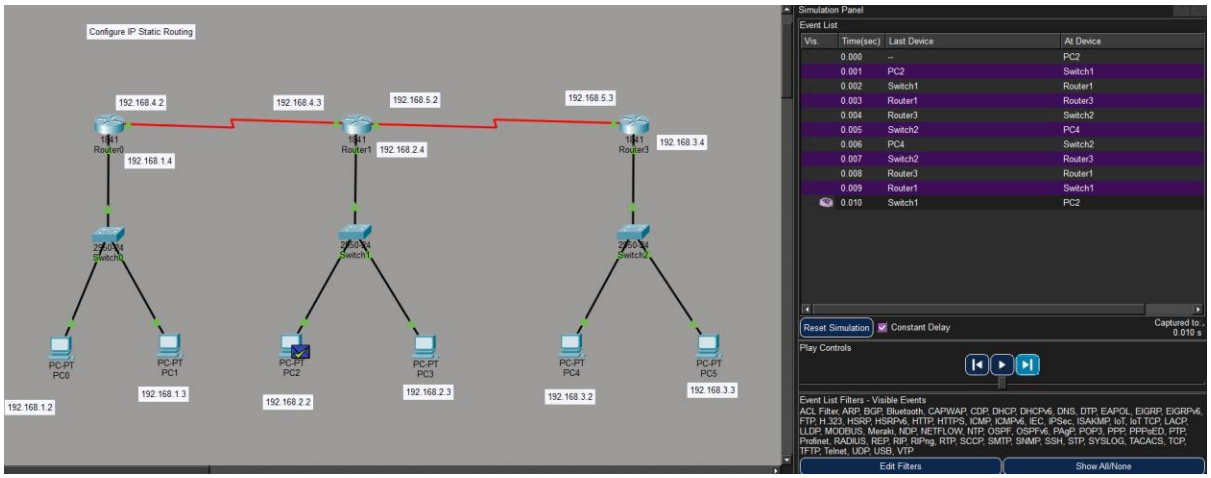

Figure 10. Packet transfer between 2 different LANs (2 and 3) successful after static routing

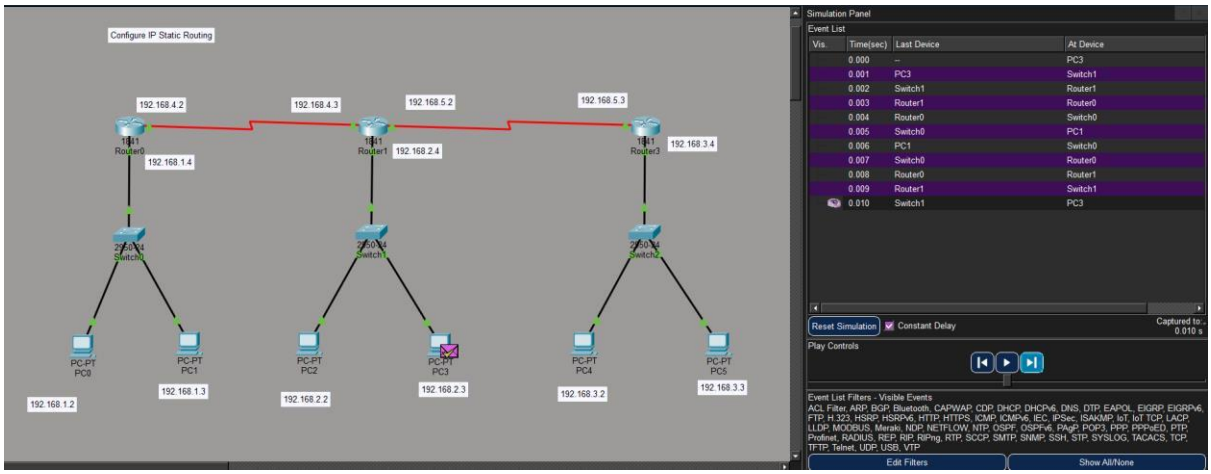

Figure 11. Packet transfer between 2 different LANs (1 and 2) successful after static routing

- After adding static routes, packets successfully traversed across multiple routers.
- The simulation event list clearly showed step-by-step forwarding:
  - PC → Switch → Router0 → Router1 → Router3 → Switch → Destination PC
- All intermediate routers were now able to **identify the correct next-hop** for each destination network.

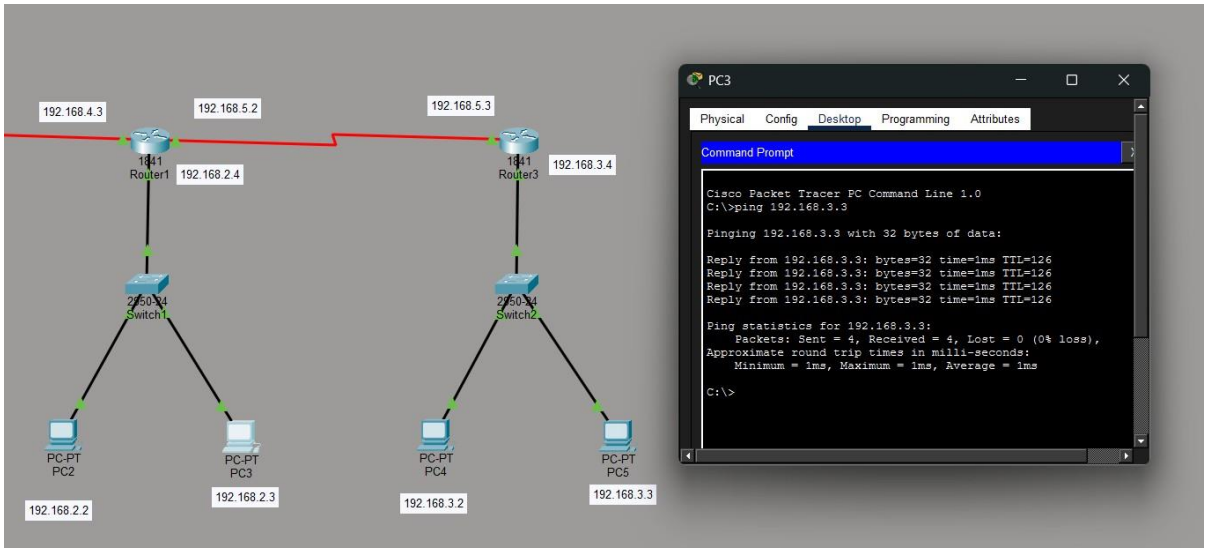

Figure 12. Ping test successful from PC3 to PC5 (2 different LANs)

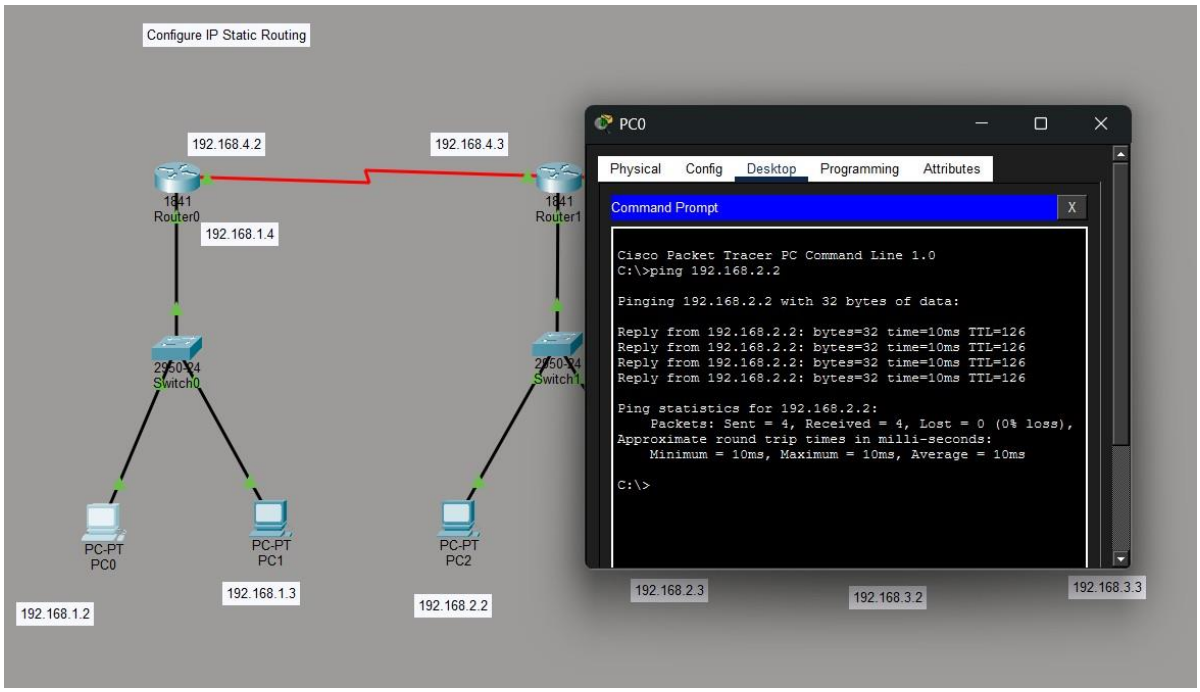

Figure 13. Ping test successful from PC0 to PC2 (2 different LANs)

- Ping tests (e.g., from PC0 to 192.168.2.2) were successful with **0% packet loss**, confirming proper connectivity.
- Round-trip time remained consistent (around 10 ms), indicating stable routing behavior.

3. Detailed Packet Flow Observation

- From the simulation traces:
  - Packets from PC0 first reached Router0, then were forwarded to Router1 using the configured next-hop (192.168.4.3).
  - Router1 forwarded packets toward Router3 using next-hop (192.168.5.3) when required.
  - Router3 delivered packets to the final LAN (192.168.3.0/24).
- Similarly, return packets followed the reverse path, confirming **bidirectional communication**.

4. Key Insights

- Static routing enabled routers to forward packets beyond their directly connected networks.

- Router1 functioned as a **central forwarding node**, handling traffic between left and right segments.
- Correct next-hop configuration is critical; any incorrect IP would result in communication failure.
- The simulation panel helped visualize packet flow and clearly demonstrated how routing decisions are applied at each hop.

Conclusion

The static routing topology was successfully configured in Cisco Packet Tracer. All routers were connected through serial links, local LANs were attached through switches, and manual routes were added to enable inter-network communication. The experiment demonstrated how static routing works and how routers forward packets based on manually configured next-hop information.

Static routing is simple and efficient for small or stable networks, but it becomes difficult to manage as the network grows because every route must be added and maintained manually.